

Fokker–Planck theory of collisions

13.1 Introduction

Logically, this chapter ought to be located at the beginning of Chapter 2, just after the discussion of phase-space concepts. This chapter is not located there because the theory in this chapter is too advanced to be so close to the beginning of the book and its location near the beginning would have delayed the introduction of other important topics that do not need the detail of this chapter.

The discussion of collisions in Chapter 1 was very approximate. Collisions were shown to scale as an inverse power of temperature, but this was based on a “one size fits all” analysis since it was assumed that collision frequencies of slow and fast particles were nominally the same as that of a particle moving at the thermal velocity. Because the collision frequency scales as v^{-3} , it is quite dubious to assume that the collision rates of both super-thermal and sub-thermal particles can be well represented by a single collision frequency and a more careful averaging over velocities is clearly warranted. This careful averaging is provided by a Fokker–Planck analysis due to Rosenbluth, Macdonald, and Judd (1957). If this much more detailed analysis simply provided more accuracy, it would not be worth the considerable effort it requires except for occasional situations where high accuracy is important. However, the Fokker–Planck theory not only provides more accuracy, but also reveals new and important phenomena and, in particular, indicates when resistive MHD fails.

We begin by reviewing collisions between two particles. Suppose a test particle having mass m_T and charge q_T collides with a field particle having mass m_F and charge q_F . Since the electric field associated with a charge q is $\mathbf{E} = \hat{r}q/4\pi\epsilon_0 r^2$, where r is the distance from the charge, the respective test and field particle equations of motion are

$$m_T \ddot{\mathbf{r}}_T = \frac{q_T q_F}{4\pi\epsilon_0 |\mathbf{r}_T - \mathbf{r}_F|^3} (\mathbf{r}_T - \mathbf{r}_F) \quad (13.1a)$$

$$m_F \ddot{\mathbf{r}}_F = \frac{q_T q_F}{4\pi\epsilon_0 |\mathbf{r}_T - \mathbf{r}_F|^3} (\mathbf{r}_F - \mathbf{r}_T). \quad (13.1b)$$

The center-of-mass vector is defined to be

$$\mathbf{R} = \frac{m_T \mathbf{r}_T + m_F \mathbf{r}_F}{m_T + m_F}. \quad (13.2)$$

Adding Eqs. (13.1a) and (13.1b) gives

$$(m_T + m_F) \ddot{\mathbf{R}} = m_T \ddot{\mathbf{r}}_T + m_F \ddot{\mathbf{r}}_F = 0 \quad (13.3)$$

showing that the center-of-mass velocity $\dot{\mathbf{R}}$ does not change as a result of the collision. If $\Delta \dot{\mathbf{r}}_T$ and $\Delta \dot{\mathbf{r}}_F$ are defined as the change in respective velocities of the two particles as result of a collision, then integration of Eq. (13.3) over the duration of the collision shows $\Delta \dot{\mathbf{r}}_T$ and $\Delta \dot{\mathbf{r}}_F$ are related by

$$(m_T + m_F) \Delta \dot{\mathbf{R}} = 0 = m_T \Delta \dot{\mathbf{r}}_T + m_F \Delta \dot{\mathbf{r}}_F. \quad (13.4)$$

It is useful to define the relative position vector

$$\mathbf{r} = \mathbf{r}_T - \mathbf{r}_F \quad (13.5)$$

and the reduced mass

$$\frac{1}{\mu} = \frac{1}{m_T} + \frac{1}{m_F}. \quad (13.6)$$

Then, dividing Eqs. (13.1a) and (13.1b) by their respective masses and taking the difference between the resulting equations gives an equation of motion for the relative velocity

$$\mu \ddot{\mathbf{r}} = \frac{q_T q_F}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}. \quad (13.7)$$

Solving Eqs. (13.2) and (13.5) for $\mathbf{r}_T$ and $\mathbf{r}_F$ gives

$$\mathbf{r}_F = \mathbf{R} - \frac{\mu}{m_F} \mathbf{r} \quad (13.8)$$

$$\mathbf{r}_T = \mathbf{R} + \frac{\mu}{m_T} \mathbf{r}$$

so the respective test and field velocities are

$$\dot{\mathbf{r}}_F = \dot{\mathbf{R}} - \frac{\mu}{m_F} \dot{\mathbf{r}} \quad (13.9)$$

$$\dot{\mathbf{r}}_T = \dot{\mathbf{R}} + \frac{\mu}{m_T} \dot{\mathbf{r}}. \quad (13.10)$$

Since $\Delta \dot{\mathbf{R}} = 0$, the change of the test and field particle velocities as measured in the lab frame can be related to the change in the relative velocity by

$$\Delta \dot{\mathbf{r}}_F = -\frac{\mu}{m_F} \Delta \dot{\mathbf{r}} \quad (13.11)$$

$$\Delta \dot{\mathbf{r}}_T = +\frac{\mu}{m_T} \Delta \dot{\mathbf{r}}. \quad (13.12)$$

The collision problem is first solved in the center-of-mass frame to find the change in the relative velocity and then the center-of-mass result is transformed to the lab frame to determine the change in the lab-frame velocity of the particles.

Let us now consider a many-particle point of view. Suppose a mono-energetic beam of particles impinges upon a background plasma. Several effects are expected to occur due to collisions between this beam and the background plasma. First, there will be a slowing down of the beam as it loses momentum due to collisions with the particles in the background plasma. Second, there should be a broadening of the beam particle velocity distribution since the collisions will also tend to randomize the velocity of the beam particles. Meanwhile, the background plasma should be heated and also should gain momentum due to the collisions. Eventually, the beam should be so slowed down and so spread out that it becomes indistinguishable from the background plasma, which will be warmer because of the energy transferred from the beam.

13.2 Statistical argument for the development of the Fokker–Planck equation

The Fokker–Planck theory (Rosenbluth, Macdonald and Judd 1957) is based on a logical argument describing how a distribution function attains its present form as a result of collisions that occurred at some earlier time. Consider a particle subject to random collisions that change its velocity. We define $F(\mathbf{v}, \Delta \mathbf{v})$ as the conditional probability that if this particle has a velocity $\mathbf{v}$ at time t , then at some later time $t + \Delta t$, collisions will have caused its velocity to become $\mathbf{v} + \Delta \mathbf{v}$.

Clearly at time $t + \Delta t$ the particle must have some velocity, so the sum of all the conditional probabilities must be unity, i.e., F must have the normalization

$$\int F(\mathbf{v}, \Delta \mathbf{v}) d\Delta \mathbf{v} = 1. \quad (13.13)$$

This definition of conditional probability can be used to show how a present distribution function $f(\mathbf{v}, t)$ comes to be the way it is because of the way it was at time $t - \Delta t$. To see how this works, consider all particles that have velocity $\mathbf{v}$ at the present time t . At some previous time $t - \Delta t$ these particles would have had velocities $\mathbf{v} - \Delta \mathbf{v}$, where $\Delta \mathbf{v}$ would have had a range of values. A particle

with some specific $\mathbf{v} - \Delta\mathbf{v}$ at the previous time would need to have undergone a collision that increased its velocity by $\Delta\mathbf{v}$ in order to get to its present state. In order to get to the present *number* of particles having velocity $\mathbf{v}$ one has to take into account how many particles had the previous velocity $\mathbf{v} - \Delta\mathbf{v}$ and then multiply this number by the probability that a $\mathbf{v} - \Delta\mathbf{v}$ particle undergoes a collision that transforms it into a particle with velocity $\mathbf{v}$. This can be expressed mathematically as

$$f(\mathbf{v}, t) = \int f(\mathbf{v} - \Delta\mathbf{v}, t - \Delta t) F(\mathbf{v} - \Delta\mathbf{v}, \Delta\mathbf{v}) d\Delta\mathbf{v}, \quad (13.14)$$

which sums up all the possible ways for obtaining a given present velocity weighted by the probability of each of these ways occurring. This analysis presumes the present status depends only on what happened during the previous collision and so is independent of all events prior to the previous collision. A partial differential equation for f can be constructed by Taylor expanding the integrand as follows:

$$\begin{aligned} f(\mathbf{v} - \Delta\mathbf{v}, t - \Delta t) F(\mathbf{v} - \Delta\mathbf{v}, \Delta\mathbf{v}) &= f(\mathbf{v}, t) F(\mathbf{v}, \Delta\mathbf{v}) - \Delta t \frac{\partial f}{\partial t} F(\mathbf{v}, \Delta\mathbf{v}) \\ &\quad - \Delta\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} (f(\mathbf{v}, t) F(\mathbf{v}, \Delta\mathbf{v})) \\ &\quad + \frac{1}{2} \Delta\mathbf{v} \Delta\mathbf{v} : \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} (f(\mathbf{v}, t) F(\mathbf{v}, \Delta\mathbf{v})). \end{aligned} \quad (13.15)$$

Substitution of Eq. (13.15) into Eq. (13.14) gives

$$\begin{aligned} f(\mathbf{v}, t) &= f(\mathbf{v}, t) \int F(\mathbf{v}, \Delta\mathbf{v}) d\Delta\mathbf{v} - \Delta t \frac{\partial f}{\partial t} \int F(\mathbf{v}, \Delta\mathbf{v}) d\Delta\mathbf{v} \\ &\quad - \int \Delta\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} (f(\mathbf{v}, t) F(\mathbf{v}, \Delta\mathbf{v})) d\Delta\mathbf{v} \\ &\quad + \frac{1}{2} \int \Delta\mathbf{v} \Delta\mathbf{v} : \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} (f(\mathbf{v}, t) F(\mathbf{v}, \Delta\mathbf{v})) d\Delta\mathbf{v}, \end{aligned} \quad (13.16)$$

where in the top line advantage has been taken of $f(\mathbf{v}, t)$ not depending on $\Delta\mathbf{v}$. Upon invoking Eq. (13.13) this can be recast as

$$\frac{\partial f}{\partial t} = \left[-\frac{\partial}{\partial \mathbf{v}} \cdot \left(\frac{\int \Delta\mathbf{v} F(\mathbf{v}, \Delta\mathbf{v}) d\Delta\mathbf{v}}{\Delta t} f(\mathbf{v}, t) \right) + \frac{1}{2} \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} : \left(\frac{\int \Delta\mathbf{v} \Delta\mathbf{v} F(\mathbf{v}, \Delta\mathbf{v}) d\Delta\mathbf{v}}{\Delta t} f(\mathbf{v}, t) \right) \right]. \quad (13.17)$$

By defining

$$\left\langle \frac{\Delta \mathbf{v}}{\Delta t} \right\rangle = \frac{\int \Delta \mathbf{v} F(\mathbf{v}, \Delta \mathbf{v}) d\Delta \mathbf{v}}{\Delta t} \quad (13.18)$$

and

$$\left\langle \frac{\Delta \mathbf{v} \Delta \mathbf{v}}{\Delta t} \right\rangle = \frac{\int \Delta \mathbf{v} \Delta \mathbf{v} F(\mathbf{v}, \Delta \mathbf{v}) d\Delta \mathbf{v}}{\Delta t} \quad (13.19)$$

the standard form of the Fokker–Planck equation is obtained,

$$\frac{\partial f}{\partial t} = -\frac{\partial}{\partial \mathbf{v}} \cdot \left(\left\langle \frac{\Delta \mathbf{v}}{\Delta t} \right\rangle f(\mathbf{v}, t) \right) + \frac{1}{2} \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} : \left(\left\langle \frac{\Delta \mathbf{v} \Delta \mathbf{v}}{\Delta t} \right\rangle f(\mathbf{v}, t) \right). \quad (13.20)$$

The first term gives the slowing down of a beam and is called the frictional term while the second term gives the spreading out of a beam and is called the diffusive term.

The goal now is to compute $\langle \Delta \mathbf{v} / \Delta t \rangle$ and $\langle \Delta \mathbf{v} \Delta \mathbf{v} / \Delta t \rangle$. To do this, it is necessary to consider all the ways collisions can cause the velocity of a specific test particle to change in a time Δt and then average all these possible values of $\Delta \mathbf{v}$ and $\Delta \mathbf{v} \Delta \mathbf{v}$ weighted according to their respective probability of occurrence. We will first evaluate $\langle \Delta \mathbf{v} / \Delta t \rangle$ and then do $\langle \Delta \mathbf{v} \Delta \mathbf{v} / \Delta t \rangle$. The problem will first be solved in the center-of-mass frame and then transformed back to the lab frame later. The center-of-mass frame analysis assumes that a particle with reduced mass μ , speed $v_{rel} = |\mathbf{v}_T - \mathbf{v}_F|$, and impact parameter b collides with a stationary scattering center located at the origin. This was discussed in Chapter 1 and the geometry was sketched in Fig. (1.3). The deflection angle θ for small-angle scattering was shown to be

$$\theta = \frac{q_T q_F}{2\pi \epsilon_0 b \mu v_{rel}^2}. \quad (13.21)$$

The averaging procedure is done in two stages:

1. The effect of collisions on a test particle in time Δt is calculated for a specific field particle velocity. The functional dependence of $\Delta \mathbf{v}$ on the scattering angle θ is determined and then used to calculate the weighted average $\Delta \mathbf{v}$ for all possible impact parameters and all possible azimuthal angles of incidence. This result is then used to calculate the weighted average change in test particle velocity in the lab frame.
2. An averaging is then performed over all possible field particle velocities weighted according to their probability, i.e., weighted according to the field particle distribution function $f_F(\mathbf{v}_F)$.

Energy is conserved in the center-of-mass frame and since the energy before and after the collision is composed entirely of kinetic energy, the magnitude of v_{rel}^2 must be the same before and after the collision. The collision therefore simply rotates the relative velocity vector by the angle θ . Let $\hat{z}$ be the direction of the relative velocity before the collision and let $\hat{b}$ be the direction of the impact parameter. If $\mathbf{v}_{rel1}, \mathbf{v}_{rel2}$ are the respective relative velocities before and after the collision, then

$$\begin{aligned}\mathbf{v}_{rel1} &= v_{rel}\hat{z} \\ \mathbf{v}_{rel2} &= v_{rel}\hat{z}\cos\theta + v_{rel}\hat{b}\sin\theta.\end{aligned}\quad (13.22)$$

Thus,

$$\begin{aligned}\Delta\mathbf{v}_{rel} &= \mathbf{v}_{rel2} - \mathbf{v}_{rel1} \\ &= v_{rel}\hat{z}(\cos\theta - 1) + v_{rel}\hat{b}\sin\theta \\ &\simeq -\frac{v_{rel}\theta^2}{2}\hat{z} + v_{rel}\theta(\hat{x}\cos\phi + \hat{y}\sin\phi),\end{aligned}\quad (13.23)$$

where the scattering angle θ is assumed to be small and ϕ is the angle between $\hat{b}$ and the x axis. The negative sign for the z component of $\Delta\mathbf{v}_{rel}$ is the fundamental reason for the drag-producing nature of collisions and is simply a consequence of the relative velocity vector rotating during the course of the collision.

The cross-section associated with impact parameter b and the range of azimuthal impact angle $d\phi$ is $bd\phi db$. In time Δt the incident particle moves a distance $v\Delta t$ and so the volume swept out for this cross-section is $bd\phi dbv\Delta t$. The number of field particles with velocity $\mathbf{v}_F$ encountered in time Δt will be the density $f_F(\mathbf{v}_F)d\mathbf{v}_F$ of these field particles multiplied by this volume, i.e., $f_F(\mathbf{v}_F)d\mathbf{v}_F bd\phi dbv\Delta t$, and so the change in relative velocity for all possible impact parameters and all possible azimuthal angles for a given $\mathbf{v}_F$ will be

$$\Delta\mathbf{v}_{|all\ b, all\ \phi} = v\Delta t f_F(\mathbf{v}_F)d\mathbf{v}_F \int d\phi \int \Delta\mathbf{v} b db. \quad (13.24)$$

The limits of integration for the azimuthal angle are from 0 to 2π and the limits of integration of the impact parameter are from the 90° impact parameter to the Debye length. On writing $\Delta\mathbf{v}$ in component form, this becomes

$$\Delta\mathbf{v}_{rel|all\ b, all\ \phi} = v_{rel}\Delta t f_F(\mathbf{v}_F)d\mathbf{v}_F \int_0^{2\pi} d\phi \int_{b_{\pi/2}}^{\lambda_D} v_{rel}(\theta\cos\phi, \theta\sin\phi, -\theta^2/2) b db. \quad (13.25)$$

The x and y integrals vanish upon integration over ϕ so

$$\begin{aligned}\Delta \mathbf{v}_{rel|all\ b, \text{ all } \phi} &= -\hat{z} v_{rel}^2 \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \pi \int_{b_{\pi/2}}^{\lambda_D} \theta^2 b db \\ &= -\hat{z} v_{rel}^2 \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \pi \int_{b_{\pi/2}}^{\lambda_D} \left(\frac{q_T q_F}{2\pi \epsilon_0 b \mu v_{rel}^2} \right)^2 b db \\ &= -\hat{z} \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \frac{q_T^2 q_F^2}{4\pi \epsilon_0^2 \mu^2 v_{rel}^2} \ln \Lambda,\end{aligned}\quad (13.26)$$

where $\Lambda = \lambda_D / b_{\pi/2}$.

The above expression can be transformed using Eq. (13.12) to give the lab-frame change in test particle velocity averaged over all impact parameters and resulting from collisions with field particles having velocity $\mathbf{v}_F$,

$$\begin{aligned}\Delta \mathbf{v}_{T|all\ b, \text{ all } \phi} &= -\hat{z} \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \frac{q_T^2 q_F^2}{4\pi \epsilon_0^2 \mu m_T v_{rel}^2} \ln \Lambda \\ &= -\Delta t \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu m_T} \frac{(\mathbf{v}_T - \mathbf{v}_F)}{|\mathbf{v}_T - \mathbf{v}_F|^3} f_F(\mathbf{v}_F) d\mathbf{v}_F.\end{aligned}\quad (13.27)$$

Next, summing over all field particle velocities gives the fully averaged change in lab-frame test particle velocity

$$\Delta \mathbf{v}_{T|all\ b, \text{ all } \phi, \text{ all } \mathbf{v}_F} = -\Delta t \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu m_T} \int \frac{\mathbf{v}_{rel}}{v_{rel}^3} f_F(\mathbf{v}_F) d\mathbf{v}_F. \quad (13.28)$$

The integrand on the right-hand side of Eq. (13.28) can be simplified using the relations

$$\frac{\partial v}{\partial \mathbf{v}} = \left(\hat{x} \frac{\partial}{\partial v_x} + \hat{y} \frac{\partial}{\partial v_y} + \hat{z} \frac{\partial}{\partial v_z} \right) \sqrt{v_x^2 + v_y^2 + v_z^2} = \frac{\mathbf{v}}{v} \quad (13.29)$$

and $\partial / \partial \mathbf{v}_T = \partial / \partial \mathbf{v}_{rel}$ to write

$$\frac{\partial v_{rel}}{\partial \mathbf{v}_T} = \frac{\mathbf{v}_{rel}}{v_{rel}} \quad (13.30)$$

and

$$\frac{\partial}{\partial \mathbf{v}_T} \frac{1}{v_{rel}} = -\frac{1}{v_{rel}^2} \frac{\mathbf{v}_{rel}}{v_{rel}}. \quad (13.31)$$

Equation (13.28) can therefore be written as

$$\Delta \mathbf{v}_{T|all\ b, \text{ all } \phi, \text{ all } \mathbf{v}_F} = \Delta t \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu m_T} \frac{\partial}{\partial \mathbf{v}_T} \int \frac{1}{v_{rel}} f_F(\mathbf{v}_F) d\mathbf{v}_F. \quad (13.32)$$

The $\partial/\partial \mathbf{v}_T$ has been factored out of the integral because $\mathbf{v}_T$ is independent of $\mathbf{v}_F$, the variable of integration. Since $\mathbf{v}_F$ is a dummy variable, it can be renamed $\mathbf{v}'$ and, since $\mathbf{v}_T$ is the velocity of the given incident particle, the subscript T can be dropped. After making these rearrangements we obtain the desired result,

$$\left\langle \frac{\Delta \mathbf{v}}{\Delta t} \right\rangle = \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu m_T} \frac{\partial}{\partial \mathbf{v}} \int \frac{f_F(\mathbf{v}')}{|\mathbf{v} - \mathbf{v}'|} d\mathbf{v}'. \quad (13.33)$$

The averaging procedure for $\Delta \mathbf{v} \Delta \mathbf{v}$ is performed in a similar manner, but instead starting with

$$\Delta \mathbf{v} \Delta \mathbf{v} = (\theta \cos \phi, \theta \sin \phi, -\theta^2/2) (\theta \cos \phi, \theta \sin \phi, -\theta^2/2), \quad (13.34)$$

in which case Eq. (13.25) is replaced by

$$\begin{aligned} \Delta \mathbf{v}_{rel} \Delta \mathbf{v}_{rel} |_{\text{all } b, \text{ all } \phi} &= v_{rel} \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \int_0^{2\pi} d\phi \int_{b_{\pi/2}}^{\lambda_D} b db \\ &\quad \times v_{rel}^2 (\theta \cos \phi, \theta \sin \phi, -\theta^2/2) (\theta \cos \phi, \theta \sin \phi, -\theta^2/2). \end{aligned} \quad (13.35)$$

The terms that are linear in $\sin \phi$ or $\cos \phi$ vanish upon performing the ϕ integration. Also, the $\hat{z}\hat{z}$ term scales as θ^4 and so may be neglected relative to the $\hat{x}\hat{x}$ and $\hat{y}\hat{y}$ terms, which scale as θ^2 . Thus, we obtain

$$\begin{aligned} \Delta \mathbf{v}_{rel} \Delta \mathbf{v}_{rel} |_{\text{all } b, \phi} &= v_{rel}^3 \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \int_0^{2\pi} d\phi \int_{b_{\pi/2}}^{\lambda_D} \theta^2 (\hat{x}\hat{x} \cos^2 \phi + \hat{y}\hat{y} \sin^2 \phi) b db \\ &= v_{rel}^3 \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F \pi (\hat{x}\hat{x} + \hat{y}\hat{y}) \int_{b_{\pi/2}}^{\lambda_D} \theta^2 b db \\ &= \Delta t f_F(\mathbf{v}_F) d\mathbf{v}_F (\hat{x}\hat{x} + \hat{y}\hat{y}) \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu^2 v_{rel}}. \end{aligned} \quad (13.36)$$

Since $\mathbf{v}_{rel}$ defines the z direction, we may write

$$\hat{x}\hat{x} + \hat{y}\hat{y} = \mathbf{I} - \frac{\mathbf{v}_{rel} \mathbf{v}_{rel}}{v_{rel}^2} \quad (13.37)$$

where $\mathbf{I}$ is the unit tensor. Thus,

$$\Delta \mathbf{v}_{rel} \Delta \mathbf{v}_{rel} |_{\text{all } b, \text{ all } \phi} = \Delta t \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu^2} f_F(\mathbf{v}_F) d\mathbf{v}_F \left(\frac{v_{rel}^2 \mathbf{I} - \mathbf{v}_{rel} \mathbf{v}_{rel}}{v_{rel}^3} \right). \quad (13.38)$$

This can be transformed using Eq. (13.12) to give

$$\Delta \mathbf{v}_T \Delta \mathbf{v}_T |_{\text{all } b, \text{ all } \phi} = \Delta t \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 m_T^2} f_F(\mathbf{v}_F) d\mathbf{v}_F \left(\frac{v_{rel}^2 \mathbf{I} - \mathbf{v}_{rel} \mathbf{v}_{rel}}{v_{rel}^3} \right). \quad (13.39)$$

The tensor may be expressed in a simpler form by noting

$$\begin{aligned}\frac{\partial}{\partial \mathbf{v}_T} \frac{\partial v_{rel}}{\partial \mathbf{v}_T} &= \frac{\partial}{\partial \mathbf{v}_T} \frac{\mathbf{v}_{rel}}{v_{rel}} \\ &= \frac{\mathbf{I}}{v_{rel}} - \frac{\mathbf{v}_{rel} \mathbf{v}_{rel}}{v_{rel}^2 v_{rel}} \\ &= \frac{v_{rel}^2 \mathbf{I} - \mathbf{v}_{rel} \mathbf{v}_{rel}}{v_{rel}^3}.\end{aligned}\quad (13.40)$$

Inserting Eq. (13.40) in Eq. (13.39) and then integrating over all the field particles gives

$$\left\langle \frac{\Delta \mathbf{v}_T \Delta \mathbf{v}_T}{\Delta t} \right\rangle = \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \int f_F(\mathbf{v}_F) d\mathbf{v}_F \frac{\partial}{\partial \mathbf{v}_T} \frac{\partial}{\partial \mathbf{v}_T} |\mathbf{v}_T - \mathbf{v}_F|. \quad (13.41)$$

However, as before, $\mathbf{v}_T$ is constant in the integrand and so $\partial/\partial \mathbf{v}_T$ may be factored from the integral. Dropping the subscript T from the velocity and renaming $\mathbf{v}_F$ to be the integration variable $\mathbf{v}'$, this can be rewritten as

$$\left\langle \frac{\Delta \mathbf{v} \Delta \mathbf{v}}{\Delta t} \right\rangle = \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} \int |\mathbf{v} - \mathbf{v}'| f_F(\mathbf{v}') d\mathbf{v}'. \quad (13.42)$$

It is convenient to define the Rosenbluth potentials

$$g_F(\mathbf{v}) = \int |\mathbf{v} - \mathbf{v}'| f_F(\mathbf{v}') d\mathbf{v}' \quad (13.43)$$

$$h_F(\mathbf{v}) = \frac{m_T}{\mu} \int \frac{f_F(\mathbf{v}')}{|\mathbf{v} - \mathbf{v}'|} d\mathbf{v}', \quad (13.44)$$

in which case

$$\begin{aligned}\left\langle \frac{\Delta \mathbf{v}}{\Delta t} \right\rangle &= \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \frac{\partial h_F}{\partial \mathbf{v}} \\ \left\langle \frac{\Delta \mathbf{v} \Delta \mathbf{v}}{\Delta t} \right\rangle &= \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \frac{\partial^2 g_F}{\partial \mathbf{v} \partial \mathbf{v}}.\end{aligned}\quad (13.45)$$

The Fokker–Planck equation, Eq. (13.20), thus becomes

$$\frac{\partial f_T}{\partial t} = \sum_{F=i,e} \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \left[-\frac{\partial}{\partial \mathbf{v}} \cdot \left(f_T \frac{\partial h_F}{\partial \mathbf{v}} \right) + \frac{1}{2} \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} : \left(f_T \frac{\partial^2 g_F}{\partial \mathbf{v} \partial \mathbf{v}} \right) \right]. \quad (13.46)$$

The first term on the right-hand side is a friction term, which describes the slowing down of the mean velocity associated with f_T while the second term is an isotropization term, which describes the spreading out (i.e., diffusive broadening) of the velocity distribution described by f_T . The right-hand side of Eq. (13.46)

thus gives the rate of change of the distribution function due to collisions and so is the correct quantity to put on the right-hand side of Eq. (2.12).

13.2.1 Slowing down

Mean velocity is defined by

$$\mathbf{u} = \frac{\int \mathbf{v} f d\mathbf{v}}{n}, \quad (13.47)$$

where $n = \int f d\mathbf{v}$. The rate of change of the mean velocity of species T is thus found by taking the first velocity moment of Eq. (13.46). Integration by parts on the right hand side terms respectively gives

$$-\int \mathbf{v} \frac{\partial}{\partial \mathbf{v}} \cdot \left(f_T \frac{\partial h_F}{\partial \mathbf{v}} \right) d\mathbf{v} = \int f_T \frac{\partial h_F}{\partial \mathbf{v}} d\mathbf{v} \quad (13.48)$$

and

$$\int \mathbf{v} \frac{\partial}{\partial \mathbf{v}} \frac{\partial}{\partial \mathbf{v}} : \left(f_T \frac{\partial^2 g_F}{\partial \mathbf{v} \partial \mathbf{v}} \right) d\mathbf{v} = - \int \frac{\partial}{\partial \mathbf{v}} \cdot \left(f_T \frac{\partial^2 g_F}{\partial \mathbf{v} \partial \mathbf{v}} \right) d\mathbf{v} = 0, \quad (13.49)$$

where Gauss' theorem has been used in Eq. (13.49). The first velocity moment of Eq. (13.46) is therefore

$$\frac{\partial \mathbf{u}_T}{\partial t} = \sum_{F=i,e} \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 n_T m_T^2} \int f_T \frac{\partial h_F}{\partial \mathbf{v}} d\mathbf{v}. \quad (13.50)$$

Let us now suppose that the test particles consist of a mono-energetic beam so that

$$f_T(\mathbf{v}) = n_T \delta(\mathbf{v} - \mathbf{u}_0). \quad (13.51)$$

In this case Eq. (13.50) becomes

$$\frac{\partial \mathbf{u}_T}{\partial t} = \sum_{F=i,e} \frac{q_T^2 q_F^2 \ln \Lambda}{4\pi \epsilon_0^2 m_T^2} \left(\frac{\partial h_F}{\partial \mathbf{v}} \right)_{\mathbf{v}=\mathbf{u}_0}. \quad (13.52)$$

Let us further suppose that the field particles have a Maxwellian distribution so that

$$f_F(\mathbf{v}) = n_F \left(\frac{m_F}{2\pi \kappa T_F} \right)^{3/2} \exp(-m_F v^2 / 2\kappa T_F) \quad (13.53)$$

and so, using Eq. (13.44),

$$h_F(\mathbf{v}) = \frac{n_F m_T}{\mu} \left(\frac{m_F}{2\pi \kappa T_F} \right)^{3/2} \int \frac{\exp(-m_F v'^2 / 2\kappa T_F)}{|\mathbf{v} - \mathbf{v}'|} d\mathbf{v}'. \quad (13.54)$$

The velocity integral in Eq. (13.54) can be evaluated using standard means (see assignments) to obtain

$$h_F(\mathbf{v}) = \frac{n_F m_T}{\mu v} \operatorname{erf} \left(\sqrt{\frac{m_F}{2\kappa T_F}} v \right), \quad (13.55)$$

where

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-w^2) dw \quad (13.56)$$

is the error function.

Thus, Eq. (13.52) becomes

$$\begin{aligned} \frac{\partial \mathbf{u}_T}{\partial t} = & \frac{n_i q_T^2 q_i^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \frac{m_T}{\mu_i} \left\{ \frac{\partial}{\partial \mathbf{v}} \left[v^{-1} \operatorname{erf} \left(\sqrt{\frac{m_i}{2\kappa T_i}} v \right) \right] \right\}_{\mathbf{v}=\mathbf{u}_0} \\ & + \frac{n_e q_T^2 q_e^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_T^2} \frac{m_T}{\mu_e} \left\{ \frac{\partial}{\partial \mathbf{v}} \left[v^{-1} \operatorname{erf} \left(\sqrt{\frac{m_e}{2\kappa T_e}} v \right) \right] \right\}_{\mathbf{v}=\mathbf{u}_0}, \end{aligned} \quad (13.57)$$

where $\mu_{i,e}^{-1} = m_{i,e}^{-1} + m_T^{-1}$.

This can be further simplified by noting (i) quasi-neutrality implies

$$n_i Z q_i + n_e q_e = 0, \quad (13.58)$$

where Z is the charge of the ions, (ii) the masses are related by

$$\frac{m_T}{\mu_{i,e}} = 1 + \frac{m_T}{m_{i,e}}, \quad (13.59)$$

and (iii) the velocity gradient of the error function must be in the direction of $\mathbf{u}_0$ using Eq. (13.29). Using these relationships and realizing that both the left and right sides are in the direction of $\mathbf{u}$, Eq. (13.57) becomes

$$\begin{aligned} \frac{\partial u}{\partial t} = & \frac{n_e e^2 \ln \Lambda}{4\pi \varepsilon_0^2} \frac{q_T^2}{m_T^2} \left[Z \left(1 + \frac{m_T}{m_i} \right) \frac{d}{du} \left(\frac{\operatorname{erf} \left(\sqrt{\frac{m_i}{2\kappa T_i}} u \right)}{u} \right) \right. \\ & \left. + \left(1 + \frac{m_T}{m_e} \right) \frac{d}{du} \left(\frac{\operatorname{erf} \left(\sqrt{\frac{m_e}{2\kappa T_e}} u \right)}{u} \right) \right]. \end{aligned} \quad (13.60)$$

Let us define

$$\xi_{i,e} = \sqrt{\frac{m_{i,e}}{2\kappa T_{i,e}}} u, \quad (13.61)$$

which is the ratio of the beam velocity to the thermal velocity of the ions or electrons. The slowing down relationship can then be expressed as

$$\begin{aligned} \frac{\partial u}{\partial t} = & \frac{n_e e^2 \ln \Lambda}{4\pi\epsilon_0^2} \frac{q_T^2}{m_T^2} \left[Z \left(1 + \frac{m_T}{m_i} \right) \frac{m_i}{2\kappa T_i} \frac{d}{d\xi_i} \left(\frac{\text{erf}(\xi_i)}{\xi_i} \right) \right. \\ & \left. + \left(1 + \frac{m_T}{m_e} \right) \frac{m_e}{2\kappa T_e} \frac{d}{d\xi_e} \left(\frac{\text{erf}(\xi_e)}{\xi_e} \right) \right]; \end{aligned} \quad (13.62)$$

the above expression describes the slowing down of a beam of mono-energetic test particles due to collisions with the electrons and ions in a background plasma.

The derivative $(\text{erf}(x)/x)'$ in Eq. (13.62) is always negative so the right-hand side of Eq. (13.62) describes a frictional drag on the test particle beam. Figure 13.1 plots $-(\text{erf}(x)/x)'$ as a function of x ; this quantity has a maximum at $x = 0.9$ indicating that frictional drag is an increasing function of x when $x < 0.9$ but when $x > 0.9$ the drag is a decreasing function of x . The $x < 0.9$ situation is consistent with ordinary notions of friction, but the $x > 0.9$ situation is like being on a “slippery slope” since the faster the particle goes, the less friction it encounters. Examination of this figure suggests $-(\text{erf}(x)/x)'$ depends linearly on x well to the left of the maximum but varies as an inverse power of x well to the right of the maximum.

This conjecture is verified by noting that Eq. (13.56) has the limiting values

$$\begin{aligned} \lim_{x \ll 1} \text{erf}(x) &\simeq \frac{2}{\sqrt{\pi}} \int_0^x (1 - w^2) dw = \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{3} \right) \\ \lim_{x \gg 1} \text{erf}(x) &\simeq 1 \end{aligned} \quad (13.63)$$

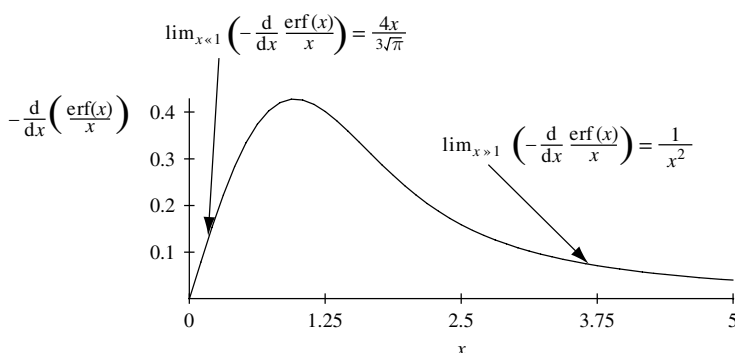


Fig. 13.1 Plot of $-(x^{-1} \text{erf}(x))'$ vs. x .

and so, as indicated in the figure,

$$\lim_{x \ll 1} \left(-\frac{d}{dx} \frac{\operatorname{erf}(x)}{x} \right) = \frac{4x}{3\sqrt{\pi}} \quad (13.64a)$$

$$\lim_{x \gg 1} \left(-\frac{d}{dx} \frac{\operatorname{erf}(x)}{x} \right) = \frac{1}{x^2}. \quad (13.64b)$$

The existence of these two different asymptotic limits and their dependence on the ratio of the test particle velocity (i.e., beam velocity) to the thermal velocity of the particles in the background plasma indicates the existence of three distinct regimes, namely:

1. Test particle is much faster than both ion and electron thermal velocities, i.e., $\xi_i, \xi_e \gg 1$.
2. Test particle is much slower than both ion and electron thermal velocities, i.e., $\xi_i, \xi_e \ll 1$.
3. Test particle is much faster than the ion thermal velocity, but much slower than the electron thermal velocity, i.e., $\xi_i \gg 1$ and $\xi_e \ll 1$.

A nominal slowing-down time can be defined by writing the generic slowing down equation

$$\frac{\partial u}{\partial t} = -\frac{u}{\tau_s} \quad (13.65)$$

so

$$\tau_s = -\frac{u}{\partial u / \partial t}. \quad (13.66)$$

This can be used to compare slowing down of test particles in the three situations listed above.

Test particle faster than both electrons, ions

Here the limit given by Eq. (13.64b) is used for both electrons and ions so that the slowing down becomes

$$\frac{\partial u}{\partial t} = -\frac{n_e e^2 \ln \Lambda}{4\pi\epsilon_0^2} \frac{q_T^2}{m_T^2} \frac{Z \left(1 + \frac{m_T}{m_i} \right) + \left(1 + \frac{m_T}{m_e} \right)}{u^2} \quad (13.67)$$

and so

$$\tau_s \simeq \frac{4\pi\epsilon_0^2}{n_e e^2 \ln \Lambda} \frac{m_T^2}{q_T^2} \frac{u^3}{Z + 1 + \frac{m_T}{m_e}} \quad (13.68)$$

since $1/m_e \gg 1/m_i$. The slowing-down time is insensitive to the plasma temperature in this case except for the weak dependence of $\ln \Lambda$ on temperature. Equation (13.67) shows that if the test particle beam is composed of electrons, the ion

friction scales as $Z/(Z+2)$ of the total friction while the electron friction scales as $2/(2+Z)$ of the total friction. In contrast, if the test particle is an ion, the friction is almost entirely from collisions with electrons. The slowing down time for ions is of the order of m_i/m_e times longer than the slowing down time for electrons having the same velocity.

Test particle slower than both electrons, ions

Here the limit given by Eq. (13.64a) is used for both electrons and ions so that the slowing-down equation becomes

$$\frac{\partial u}{\partial t} = -\frac{u}{3\sqrt{\pi}} \frac{n_e e^2 \ln \Lambda}{\pi \epsilon_0^2} \frac{q_T^2}{m_T^2} \left(Z \left(1 + \frac{m_T}{m_i} \right) \left(\frac{m_i}{2\kappa T_i} \right)^{3/2} + \left(1 + \frac{m_T}{m_e} \right) \left(\frac{m_e}{2\kappa T_e} \right)^{3/2} \right) \quad (13.69)$$

and the slowing-down time becomes

$$\tau_s = \frac{3\sqrt{\pi} \pi \epsilon_0^2 m_T^2}{n_e e^2 \ln \Lambda} \frac{1}{q_T^2 \left(Z \left(1 + \frac{m_T}{m_i} \right) \left(\frac{m_i}{2\kappa T_i} \right)^{3/2} + \left(1 + \frac{m_T}{m_e} \right) \left(\frac{m_e}{2\kappa T_e} \right)^{3/2} \right)}. \quad (13.70)$$

The temperature terms scale as the inverse cube of the thermal velocity and so if the ion and electron temperatures are the same order of magnitude, then the ion contribution dominates. Thus, the slowing down is mainly done by collisions with ions and the slowing-down time is

$$\tau_s \simeq \frac{3\sqrt{\pi} \pi \epsilon_0^2 m_T^2}{n_e e^2 \ln \Lambda} \frac{(2\kappa T_i/m_i)^{3/2}}{q_T^2 Z \left(1 + \frac{m_T}{m_i} \right)}. \quad (13.71)$$

The slowing down of a very slow beam is thus temperature-sensitive and, in particular, this sort of beam takes a long time to slow down in hot plasmas.

Intermediate case: beam faster than ions, slower than electrons

In this case, the slowing-down equation becomes

$$\frac{\partial u}{\partial t} = -\frac{n_e e^2 \ln \Lambda}{4\pi \epsilon_0^2} \frac{q_T^2}{m_T^2} \left(Z \left(1 + \frac{m_T}{m_i} \right) \frac{1}{u^2} + \left(1 + \frac{m_T}{m_e} \right) \left(\frac{m_e}{2\kappa T_e} \right)^{3/2} \frac{4u}{3\sqrt{\pi}} \right) \quad (13.72)$$

and the slowing-down time is

$$\tau_s = \frac{4\pi\epsilon_0^2 m_T^2}{n_e q_T^2 e^2 \ln \Lambda \left(\left(1 + \frac{m_T}{m_i}\right) \frac{Z}{u^3} + \frac{4}{3\sqrt{\pi}} \left(1 + \frac{m_T}{m_e}\right) \left(\frac{m_e}{2\kappa T_e}\right)^{3/2} \right)}. \quad (13.73)$$

13.3 Electrical resistivity

A uniform steady electric field imposed on a plasma accelerates ions and electrons in opposite directions. The accelerated particles will collide with other particles and this frictional drag will oppose the acceleration. A steady state might be achieved where the accelerating force due to the electric field balances the drag force due to collisions. Because the electric field causes equal and opposite momentum gains by the electrons and ions it does not alter the center-of-mass momentum of the entire plasma. Furthermore, electron–electron collisions cannot change the average momentum of the electrons nor can ion–ion collisions change the average momentum of the ions. The only way for the average electron momentum to change is by collisions with ions and vice versa. Thus, electrical resistivity must only depend on collisions between electrons and ions because electrical resistivity is a relation involving current, which is a function that depends on average electron momentum and average ion momentum.

We postulate that this balance between acceleration and drag results in an equilibrium where the electrons and ions have shifted Maxwellian distribution functions

$$\begin{aligned} f_i(\mathbf{v}) &= \frac{n_i}{\pi^{3/2} (2\kappa T_i / m_i)^{3/2}} \exp \left(-m_i (\mathbf{v} - \mathbf{u}_i)^2 / 2\kappa T_i \right) \\ f_e(\mathbf{v}) &= \frac{n_e}{\pi^{3/2} (2\kappa T_e / m_e)^{3/2}} \exp \left(-m_e (\mathbf{v} - \mathbf{u}_e)^2 / 2\kappa T_e \right). \end{aligned} \quad (13.74)$$

Since $\mathbf{u}_i$ and $\mathbf{u}_e$ are the respective average ion and electron velocities, the current density is

$$\mathbf{J} = n_i q_i \mathbf{u}_i + n_e q_e \mathbf{u}_e. \quad (13.75)$$

It is convenient to transform to a frame moving with the electrons, i.e., with the velocity $\mathbf{u}_e$. In this frame the electron mean velocity will be zero and the ion mean velocity will be $\mathbf{u}_{rel} = \mathbf{u}_i - \mathbf{u}_e$. The current density remains the same because it is proportional to $\mathbf{u}_{rel}$, which is frame-independent. The distribution functions in this frame will be

$$f_i(\mathbf{v}) = \frac{n_i}{\pi^{3/2} (2\kappa T_i / m_i)^{3/2}} \exp \left(-m_i (\mathbf{v} - \mathbf{u}_{rel})^2 / 2\kappa T_i \right)$$

$$f_e(\mathbf{v}) = \frac{n_e}{\pi^{3/2}(2\kappa T_e/m_e)^{3/2}} \exp(-m_e v^2/2\kappa T_e). \quad (13.76)$$

Since the ion thermal velocity is much smaller than the electron thermal velocity, the ion distribution function is much narrower than the electron distribution function and the ions can be considered as constituting a mono-energetic beam impinging upon the electrons. Using Eq. (13.57), the net force on this ion beam due to the combination of frictional drag on electrons and acceleration due to the electric field is

$$m_i \frac{\partial \mathbf{u}_{rel}}{\partial t} = \frac{n_e q_i^2 q_e^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu_e} \left\{ \frac{\partial}{\partial \mathbf{v}} \left[v^{-1} \operatorname{erf} \left(\sqrt{\frac{m_e}{2\kappa T_e}} v \right) \right] \right\}_{\mathbf{v}=\mathbf{u}_{rel}} + q_i \mathbf{E}. \quad (13.77)$$

In steady state the two terms on the right-hand side balance each other, in which case

$$\mathbf{E} = -\frac{n_e q_i e^2 \ln \Lambda}{4\pi \epsilon_0^2 \mu_e} \left\{ \frac{\partial}{\partial \mathbf{v}} \left[v^{-1} \operatorname{erf} \left(\sqrt{\frac{m_e}{2\kappa T_e}} v \right) \right] \right\}_{\mathbf{v}=\mathbf{u}_{rel}}. \quad (13.78)$$

If u_{rel} is much smaller than the electron thermal velocity, then Eq. (13.64a) can be used to obtain

$$\mathbf{E} = \frac{n_e q_i e^2 \ln \Lambda}{3\sqrt{\pi} \pi \epsilon_0^2 m_e} \frac{\mathbf{u}_{rel}}{(2\kappa T_e/m_e)^{3/2}} = \frac{Ze^2 m_e^{1/2} \ln \Lambda}{3\sqrt{\pi} \pi \epsilon_0^2} \frac{\mathbf{J}}{(2\kappa T_e)^{3/2}}, \quad (13.79)$$

where $\mathbf{u}_{rel} = \mathbf{J}/n_e e$ and $\mu_e^{-1} = 1/m_i + 1/m_e \simeq 1/m_e$ have been used. Equation (13.79) can be used to define the electrical resistivity

$$\eta = \frac{Ze^2 m_e^{1/2}}{3\sqrt{\pi} \pi \epsilon_0^2} \frac{\ln \Lambda}{(2\kappa T_e)^{3/2}} \quad (13.80)$$

so that

$$\mathbf{E} = \eta \mathbf{J}. \quad (13.81)$$

If the temperature is expressed in terms of electron volts then $\kappa = e$ and the resistivity is

$$\eta = 1.03 \times 10^{-4} \frac{Z \ln \Lambda}{T_e^{3/2}} \text{ Ohm m}. \quad (13.82)$$

This resistivity is more accurate than the rough calculation given by Eq. (1.25); the numerical coefficient here is smaller by about a factor of three.

13.4 Runaway electric field

The evaluation of the error function in Eq. (13.78) involved the assumption that u_{rel} , the relative drift between the electron and ion mean velocities, is much

smaller than the electron thermal velocity. This assumption implies $-(x^{-1} \operatorname{erf}(x))' \simeq 4x/3\sqrt{\pi}$, so x must lie well to the left of 0.9 in the curve plotted in Fig. 13.1. Consideration of Fig. 13.1 shows that $-(x^{-1} \operatorname{erf}(x))'$ has a maximum value of 0.43, which occurs when $x = 0.9$ so that Eq. (13.78) cannot be satisfied if

$$E > \max \left\{ -\frac{n_e q_i e^2 \ln \Lambda}{4\pi \epsilon_0^2 m_e} \frac{\partial}{\partial v} \left[v^{-1} \operatorname{erf} \left(\sqrt{\frac{m_e}{2\kappa T_e}} v \right) \right] \right\} = 0.43 \frac{n_e Z e^3 \ln \Lambda}{8\pi \epsilon_0^2 \kappa T_e}. \quad (13.83)$$

When the electric field is large enough to satisfy this inequality, the entire concept of electrical resistivity fails (Dreicer 1959) because the underlying presumption of the resistivity calculation, namely a steady-state balance between drag and electric field acceleration, becomes false. If the electron temperature is expressed in terms of electron volts, attaining a balance between the acceleration due to the electric field and the deceleration due to frictional drag thus becomes impossible if

$$E > E_{\text{Dreicer}}, \quad (13.84)$$

where the Dreicer electric field is defined by

$$\begin{aligned} E_{\text{Dreicer}} &= 0.43 \frac{n_e Z e^3 \ln \Lambda}{8\pi \epsilon_0^2 \kappa T_e} \\ &= 5.6 \times 10^{-18} n_e Z \frac{\ln \Lambda}{T_e} \text{ V/m}. \end{aligned} \quad (13.85)$$

If the electric field exceeds E_{Dreicer} then the frictional drag lies to the right of the maximum in Fig. 13.1. No equilibrium is possible in this case as can be seen by considering the sequence of collisions of a nominal particle. The acceleration due to E between collisions causes the particle to go faster, but since it is to the right of the maximum, the particle has less drag when it goes faster. If the particle has less drag, then it will have a longer mean free path between collisions and so be accelerated to an even higher velocity. The particle velocity will therefore increase without bound if the system is infinite and uniform. In reality, the particle might exit the system if the system is finite or it might radiate energy. Very high, even relativistic velocities can easily develop in these runaway situations.

This runaway analysis is important to the characterization of magnetic reconnection. In particular, if the electric field in a reconnecting plasma satisfies Eq. (13.84), then the equation $\mathbf{E} = \eta \mathbf{J}$ cannot be used, in which case resistive models of reconnection become inappropriate and collisionless models must be used.

13.5 Assignments

1. Evaluate the integral in Eq. (13.54) and show that it leads to Eq. (13.55). Hint: since $\mathbf{v}$ is a fixed parameter in the integral, let the direction of $\mathbf{v}$ define the axis of a spherical polar coordinate system. Let $\boldsymbol{\xi} = \mathbf{v}' - \mathbf{v}$ and let θ be the angle between $\boldsymbol{\xi}$ and $\mathbf{v}$. Then note that $(v')^2 = \xi^2 + 2\boldsymbol{\xi} \cdot \mathbf{v} + v^2$ and $d\mathbf{v}' = d\boldsymbol{\xi}$. Express $d\boldsymbol{\xi}$ in spherical polar coordinates and let $x = \cos \theta$.
2. Interaction of a low-density, fast electron beam with a cold background plasma. Assume that the velocity of a beam of electrons impinging on a plasma is much faster than the velocities of the background electrons and ions. The ions have charge Z so that, ignoring the beam density, the quasi-neutrality condition is $Zn_i = n_e$.
 - (a) Show that the Rosenbluth h potential can be approximated as

$$h_F(\mathbf{v}) = \frac{m_T}{\mu} \int \frac{f_F(\mathbf{v}')}{|\mathbf{v} - \mathbf{v}'|} d\mathbf{v}'$$

$$\simeq \frac{m_e n_F}{v\mu},$$

where

$$\frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_F}.$$

What is the form of $h_F(\mathbf{v})$ for beam electrons interacting with background plasma

(i) electrons and (ii) ions?

- (b) By approximating

$$(v^2 - 2\mathbf{v} \cdot \mathbf{v}' + v'^2)^{1/2} = v \left(1 - \frac{\mathbf{v} \cdot \mathbf{v}'}{v^2} + \frac{v'^2}{2v^2} \right),$$

where the beam velocity v is much larger than the background species velocity v' , show that the Rosenbluth g potential can be approximated as

$$g_F(\mathbf{v}) = \int |\mathbf{v} - \mathbf{v}'| f_F(\mathbf{v}') d\mathbf{v}'$$

$$\simeq \left(v + \frac{1}{v} \frac{3\kappa T_F}{2m_F} \right) n_F.$$

Hint: use symmetry arguments when considering the term involving $\mathbf{v} \cdot \mathbf{v}'$.

- (c) Assume the beam velocity is z -directed and let $F_T = \int d^2v_\perp f_T$ so F_T describes the projection of the 3-D beam distribution onto the z axis of velocity space. Since $v = v_z$ and $v_y = v_x \approx 0$ show it is possible to write

$$h_F(\mathbf{v}) \simeq \frac{m_e n_F}{v_z \mu}$$

$$g_F(\mathbf{v}) \simeq \left(v_z + \frac{1}{v_z} \frac{3\kappa T_F}{2m_F} \right) n_F.$$

Show that the Fokker–Planck equation can be written as

$$\begin{aligned} \frac{\partial F_T}{\partial t} &\simeq \sum_{F=i,e} \frac{n_F q_T^2 q_F^2 \ln \Lambda}{4\pi\epsilon_0^2 m_T^2} \left[\frac{\partial}{\partial v_z} \left(F_T \frac{m_T}{\mu v_z^2} \right) + \right. \\ &\quad \left. \frac{1}{2} \frac{\partial^2}{\partial v_z^2} \left(F_T \frac{\partial^2}{\partial v_z^2} \left(v_z + \frac{3\kappa T_F}{2m_F v_z} \right) \right) \right] \\ &\simeq \sum_{F=i,e} \frac{n_F q_T^2 q_F^2 \ln \Lambda}{4\pi\epsilon_0^2 m_T^2} \frac{\partial}{\partial v_z} \left[F_T \frac{m_T}{\mu v_z^2} + \frac{3}{2v_z^3} \frac{\kappa T_F}{m_F} \frac{\partial F_T}{\partial v_z} \right]. \end{aligned}$$

(d) Taking into account charge neutrality and $v_z \gg v_{Ti}$ show this can be recast as

$$\frac{\partial F_T}{\partial t} \simeq \frac{n_e e^4 \ln \Lambda}{4\pi\epsilon_0^2 m_e^2} \frac{\partial}{\partial v_z} \left[F_T \frac{2+Z}{v_z^2} + \frac{3}{2v_z^3} \frac{\kappa T_e}{m_e} \frac{\partial F_T}{\partial v_z} \right].$$

(e) Show that a steady-state equilibrium can develop where, because of collisions with background electrons and ions, the fast beam distribution has the form

$$F_T \sim \exp \left(-\frac{(2+Z)m_e v_z^2}{3\kappa T_e} \right).$$

Is this consistent with the original assumption that the beam is fast compared to the background plasma?

3. An axisymmetric plasma has a magnetic field that can be expressed as

$$\mathbf{B} = \frac{1}{2\pi} (\nabla\psi \times \nabla\phi + \mu_0 I \nabla\phi),$$

where ϕ is the toroidal angle, $\psi(r, z)$ is the poloidal flux, and I is the current flowing through a circle of radius r at axial position z .

(a) Show that the toroidal component of the vector potential is

$$A_\phi(r, z, t) = \frac{1}{2\pi r} \psi(r, z, t).$$

(b) Assume that the plasma obeys the resistive Ohm's law

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}$$

and assume that the plasma is stationary so that the toroidal component is simply

$$E_\phi = \eta J_\phi$$

and use Eq. (9.42) to show that

$$J_\phi = -\frac{r}{2\pi\mu_0} \nabla \cdot \left(\frac{1}{r^2} \nabla\psi \right)$$

so that the toroidal component of Ohm's law is

$$E_\phi = -\frac{r\eta}{2\pi\mu_0} \nabla \cdot \left(\frac{1}{r^2} \nabla\psi \right).$$

- (c) Assuming classical resistivity $\eta \sim T_e^{-3/2}$ sketch the temperature dependence of $|E_\phi|$ as given above and also sketch the temperature dependence of $E_{Dreicer}$ as given by Eq. (13.85). For a plasma with given ψ and physical dimensions, at what electron temperature limit do runaway electrons develop (high or low temperature)? For a given temperature, do runaways develop with high ψ or low ψ ? For a given temperature and flux, do runaways develop in a large device or in a small device? If the plasma density decays, will runaways develop?
4. Discuss qualitatively (no mathematics) what would happen if the electric field at the X-point of the Sweet–Parker reconnection configuration discussed in Assignment 1 of Chapter 12 exceeded $E_{Dreicer}$. By taking into account azimuthal symmetry discuss what constraints exist for the canonical angular momentum of a particle in the vicinity of the X-point. Discuss the particle orbits taking into account the results of Assignment 2 of Chapter 3. How might the particle motion relate to the excitation of waves and what sort of waves might be excited? How might this constitute an anomalous resistivity?